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REGULAR PAPER

Investigation of graph-based clustering approaches along with graph neural networks for modeling armed conflict in Bangladesh

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Abstract
Determining fatality rates—a critical component of conflict analysis and comprehending the dynamics of armed conflict in Bangladesh are the main goals of this study. The contributions of this paper are twofold. Firstly, this paper presents the performance of graph neural networks (GNNs) is measured against traditional machine learning (ML) algorithms, specifically artificial neural networks (ANNs), for fatality prediction. Secondly, it has presented an empirical analysis of graph-based clustering and traditional clustering methods for modeling armed conflicts in Bangladesh. Predictive fatality models have been developed using graph neural networks and traditional machine learning algorithms. Graph neural networks exploit the graph structure of the data, allowing for more effective modeling and prediction. Traditional machine learning algorithms, and artificial neural networks, serve as benchmarks for comparison. The results indicate that graph-based clustering methods provide valuable insights into the structure and dynamics of armed conflicts in Bangladesh. The comparative analysis demonstrates that graph neural networks outperform a few traditional machine learning algorithms in terms of predictive fatality, highlighting the importance of capturing the relational dependencies present in the data. Graph-based clustering techniques offer an innovative approach by capturing the inherent relational structure of conflicting data, whereas traditional clustering methods rely on traditional feature engineering and distance metrics. A comprehensive dataset from ACLED for armed conflict incidents in Bangladesh was used to evaluate the performance. Both graph-based and traditional clustering methods were applied to identify meaningful clusters and analyze the patterns of armed conflict. An additional contribution has been made through the construction and analysis of a knowledge graph, elucidating the interconnected relationships among various datasets. In conclusion, this research contributes to the field of conflict analysis by showcasing the advantages of graph-based clustering methods and graph neural networks for understanding armed conflict patterns and predicting fatality rates.

Keywords Conflict events · Fatalities · Armed conflict · Hierarchical clustering · Machine learning · Spatial analysis · Predictive modeling

1 Introduction

Bangladesh is a densely populated developing country in South Asia. According to the Global Organized Crime Index (https://ocindex.net/country/bangladesh), Bangladesh ranked 89th and 126th in criminality index and resilience score among 193 countries. The best rank Bangladesh has ever achieved in world peace parameters is 82. These statistics show that Bangladesh is not among the peaceful countries. One of the primary reasons behind it is internal armed conflict. We aim to find the hidden factors that fuel

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these armed conflicts [1]. We try to answer questions such as "What to expect when different actors are involved in the conflict? "What regions are impacted the most?" "What are the events that manifest more fatalities?" and "Can we model and predict fatalities based on the features we have?"

In the literature, various studies have been featured to demonstrate the complexity of armed conflict, and the range of effects it has on society. According to Roberts et al., the main causes of trauma and armed conflict-related deaths and injuries globally include natural catastrophes, armed conflicts, and low-resource countries with inadequate health systems. Negret et al. [7] investigated the poorly recognized connection between deforestation and armed conflict, emphasizing the role played by illicit agricultural cultivation, especially in biodiverse zones. Badiuzzaman et al. [6], for example, focused on rural household livelihood decisions in post-war areas like Bangladesh's Chittagong Hill Tracts. Furthermore, using key indicators including population size, infant mortality rates, and education levels, predictive model predicted changes in armed conflict both globally and regionally, showing a decrease in the number of nations experiencing internal armed conflict. However, the studies cited above underscore the complex and varied effects that armed conflict has on society and the environment, highlighting the need for armed conflict assessment to take a wide variety of issues into account. There has been a noticeable dearth of thorough research on the underlying causes of armed conflict, despite the fact that a lot of study has been done on the prediction of crime kinds and patterns. Prior research has tended to ignore the promise of graph-based techniques, preferring regression models and conventional machine learning algorithms for prediction. While several researches have used clustering approaches, the potential of graph-based clustering algorithms has not been sufficiently explored. Moreover, the previous studies have frequently failed to look at regional variations in the dynamics of armed conflict as well as spatial patterns.

Graph-based approaches can provide more contextually rich and nuanced clustering than standard methods that only use density or distance metrics. Due to its capacity to capture intricate links and dependencies among datasets, graph-based clustering offers notable advantages in a number of disciplines. This is because graph-based approaches take advantage of the intrinsic structure of the data they represent. Graph-based clustering looks at the relationships and interactions between data points to find hidden patterns and communities that conventional grouping techniques can miss. This holds significant value for handling high-dimensional and sparse datasets or in situations where data display nonlinear connections. However, a family of deep learning models called graph neural networks (GNNs) is intended for processing and analyzing structured data that are represented as graphs. They facilitate efficient learning and inference tasks by acting directly on the graph structure. GNNs are useful for tasks such as node classification, link prediction, and graph classification because they can recognize intricate connections and patterns in graph-structured data by utilizing message-passing and aggregation techniques.

To address all the limitations, we have contributed in two ways: to forecast fatalities, this study first compares the effectiveness of graph neural networks (GNNs) with more conventional machine learning (ML) algorithms, namely, artificial neural networks (ANNs), and to model violent conflicts in Bangladesh, it has provided an empirical comparison between standard clustering approaches and graph-based clustering. At first, we conducted a comparative study of the prediction abilities of artificial neural networks (ANN), graph-based neural networks (GNN), and conventional machine learning (ML) algorithms. Our results show that GNN performs similarly to RandomForest and AdaBoost Regressor Methods, outperforming KNearestRegressor and ANN. This highlights the promise of using graph-based approaches for predictive modeling, even though some ML-based algorithms perform better than GNN. We use graph-based clustering methods to group actors according to the interactions that take place between them. We seek to optimize the clustering result by experimenting with several graph-based clustering techniques, and we finally employ ensemble clustering to achieve our goal. Furthermore, we employ clustering algorithms to pinpoint regions according to the frequency of fatalities. Moreover, by linking players and events to every cluster, we reveal hidden characteristics. Since fatality rates define clusters, we can deduce the kind of crimes and players that may be involved in areas with the greatest death rates.

Our research offers a significant contribution through a comparative analysis of the effectiveness between graph-based and conventional clustering techniques, along with an examination of the predictive capability of graph neural networks (GNNs) in forecasting fatalities. Additionally, we provide actionable recommendations aimed at preventing and resolving conflicts. Our findings furnish valuable insights for policy-makers and humanitarian organizations, focusing on the understanding of relational patterns and underlying causes to effectively address violent conflicts in Bangladesh. Moreover, by harnessing the potential of graph-based methodologies, our research introduces novel approaches for modeling and predicting violent conflicts, thus offering new avenues for intervention and conflict resolution strategies. The three major contributions of this study are as follows:

- This study evaluates the efficacy of graph neural networks (GNNs) in comparison with traditional machine learning

algorithms for predicting fatalities. It highlights the capability and effectiveness of GNNs in capturing relational dependencies, showcasing their potential for improved performance in this predictive task.

- This research compares classic clustering techniques with graph-based clustering to describe armed conflicts in Bangladesh. It highlights the efficiency of graph-based techniques in representing relational structures and provides insightful information on why they are better to traditional techniques in this respect.
- This research addresses data visualization easier, and a knowledge graph has been created based on armed conflict. It offers a thorough foundation for comprehending the complex interactions and linkages included in the armed conflict dataset.
- In addition to giving legislators and humanitarian organizations new perspectives on how to confront violent conflicts in Bangladesh, the research's conclusions have practical consequences for preventing and resolving conflicts. The aforementioned discoveries provide relevant parties with updated knowledge and tactics to efficiently tackle and alleviate issues associated with conflicts within the area.

2 Literature review

To understand the origins and consequences of armed conflict, several studies have been carried out. This kind of system benefits from a number of fascinating properties provided by machine learning. It is important to evaluate the limits of these investigations seriously even when they have yielded interesting insights. Our findings about the technology employed in the previous works and their limitations are compiled in Table 1. Our investigation's tight ties to the research were the only ones we shortlisted.

Along with the studies mentioned in Table 1, the following studies show that armed conflict can depend on various factors and have different impacts on society. We keep this section to show the variation of aspects we should consider while considering armed conflicts. SE Roberts et el. [9] investigated low-resource nations with weak health systems, armed wars, and natural disasters as major causes of deaths and injuries in armed conflict and trauma worldwide [27]. This research [10] examined the hidden but ongoing cost of conflict to birth weight outcomes in 53 developing countries that have experienced conflict in the preceding 30 years (1990–2018) and showed that armed conflict in the first trimester reduces birth weight by 2.8% and increases low birth weight by 3.2 percentage points. The influence of armed conflict on deforestation in biodiverse regions of the planet remains poorly understood. Its relationship with illegal agricultural cultivation can hide its effects on deforestation patterns [11]. This series of studies examined the direct and indirect health effects of armed conflict on women and children (including adolescents) worldwide [3]. This study by Badiuzzaman et al. [12] examines rural household livelihood decisions, including prospective educational investment in the post-conflict Chittagong Hill Tracts (CHT) region of Bangladesh. The article by Herge [13] predicted armed conflict changes globally and regionally for 2010–2050, using a dynamic multinomial logit model with core predictors such as population size, infant mortality rates, and education levels. Predictions indicate a decline from 15 to 7% for countries experiencing internal armed conflict. As these studies show that armed conflict can depend on different types of features, we used several features in consideration while predicting results.

Global peace is a crucial aspect of international relations and human prosperity with the aim of ensuring the absence of armed conflicts, violence, and tensions among nations and communities. The following studies show how the world is trying to reduce this violence. For decades, international societies have attempted to create comprehensive, accurate, and useful early warning systems for political and local disturbances [14]. To reduce conflicts, the UN promotes [15] social progress, human rights, world peace, and peacekeeping cite wars.

The intersection of deep learning and various domains has led to significant advancements in predictive modeling. Prakash et al. [16] introduced an innovative approach integrating evolutionary feature selection and ensemble methods for cardiovascular disease prediction. Their work demonstrated enhanced predictive performance by optimizing feature selection and leveraging ensemble techniques. In the realm of sentiment analysis, Paul et al. [17] proposed a context-sensitive multi-tier deep learning framework for multimodal sentiment analysis, emphasizing the nuanced understanding of sentiment across different contexts. Prakash et al. [18] extended their exploration into health care by developing dual-layer deep ensemble techniques for heart disease classification. Their study, highlighted the efficacy of leveraging deep ensemble methods for accurate disease classification, thus aiding in timely diagnosis and treatment. Vijai et al. [19] proposed a modified firefly deep ensemble for microarray data classification, focusing on improving classification accuracy and interpretability in biological data analysis. Their work demonstrated advancements in classifying complex biological datasets. Subramanian et al. [20] introduced a deep ensemble network model for breast cancer classification and prediction. Their research showcased the potential of deep ensemble techniques in improving cancer diagnosis accuracy and prognostic capabilities. Jothi et al. [21] presented a multi-aspect framework for explainable sentiment

Table 1 Previous research works in terms of objectives, used techniques, and possible research gaps

References	Research purpose	Used techniques	Focused methods	Challenges/research gaps
[1]	A machine learning approach to predict crime using time and location data	Decision Tree, Naive Bayesian, Logistic Regression, k-Nearest Neighbor, Ensemble Method, and SMOTE for imbalance data	Focused on geographical location basic details to predict a type of crime using supervised classification methods	The absence of actors related to crime, poor accuracy over all the methods, Only under-sampling brings 81% accuracy, clustering is also missing
[2]	Aims to scrutiny the nature of armed conflict and terrorism in South Asia	Qualitative and quantitative analysis	Conducted a statistical analysis of the MIPT datasets and found that armed conflict and terrorism increase rapidly in lagging regions in comparison with the leading region of South Asia	Does not directly address the policies of the South Asian countries but rather uses already available literature of policy experts to conclusively make a judgment
[3]	Bangladesh Crime Reports Analysis and Prediction	Decision Tree, Random Forest, Lasso, and MLP. Ridge, SVR, and Linear Regression	Focused on predicting, preventing, and solving future crimes, outputting the coordinates first in the first model (regression) then using that output as input in the second model (classification), and then predicting the event type (MLP)	Lacking of predicting fatalities. Actors involved in clashes, 0.060 is the score of MLP which indicates that it fails to recognize event types
[4]	Using Linear Regression to Forecast Future Trends in Crime of Bangladesh	Linear Regression	Predicted forecasted murders using Linear Regression	Did not use other regression methods and no clear sign of why the dataset is linear in all the cases and how to capture nonlinearity, only focused in one at a tribute (murder number)
[5]	Forecasting fatalities in armed conflict up to March 2025	A number of smaller forecasting models—models and ensemble ML methods	Offers political Violence Early Warning System (ViEWS) model to predict future fatality	The key challenge of this research is it demands a violent record for a long period. When forecasting 36 months into the future. The model expectantly performed less well
[6]	Visualization and prediction of Boston crime data	R. Decision Tree and Random Forest	Focused on foretelling crimes in Boston city	Their dataset contains complex crime distribution dataset; hence, they achieve lower accuracy
[7]	Identifying and foreseeing domestic political crisis	Interaction graph. Subgraph matching algorithm, basic machine learning model 5	Focused on a graph-based approach for detecting and forecasting domestic crisis. For this, they constructed an interaction graph from the dataset to model the domestic interactions and found the feature vector using a subgraph matching algorithm. For forecasting, they employed a basic machine learning model	The dataset they worked with is pretty small to make predictions

Table 1 (continued)

References	Research purpose	Used techniques	Focused methods	Challenges/research gaps
[8]	Dyad Classification for Military Conflict Analysis	Textual and graph-based attributes, substantial graph	Put a focus on determining the conflict between two entities. To show a cause and qualities portrayed as a considerable graph. A collection of textual and graph-based features is used. Classifiers are suggested for determining whether a particular pair of entities are allies or adversaries	Only analyze the correlated conflict between two entities (allies or on emics)

analysis, aiming to provide deeper insights into the sentiment analysis process and enhance model interpretability. Their work emphasized the importance of transparency and interpretability in sentiment analysis applications. Hossain et al. [22] introduced Smart-Agri, an integrated framework leveraging IoT, machine learning, and blockchain for smart agricultural management. Rahaman et al. [23] developed a deep learning-based smartphone application for detecting mango diseases and providing pesticide suggestions. Rahman et al. [24] explored the application of convolutional neural networks (CNNs) for multi-object recognition from RGB images. By leveraging deep learning techniques, their work contributes to advancements in computer vision, offering promising avenues for enhancing object recognition systems in various real-world applications. Venugopal et al. [25] underscored the burgeoning interest and potential of immersive virtual environments. Their comprehensive review delves into various facets of the metaverse, spanning technological advancements, user experiences, social interactions, and potential applications across industries.

Our research endeavors to positively impact Bangladesh's position in the global peace landscape and create a more peaceful and harmonious society.

3 Research gap

The potential of graph-based approaches has frequently been underestimated in the previous research, which has favored traditional machine learning algorithms and regression models for predictive tasks. To rectify this mistake, our study compares the ability of artificial neural networks (ANN), graph-based neural networks (GNN), and conventional machine learning (ML) techniques to predict fatalities. Our results show that GNN performs better than KNearestRegressor and ANN and is competitive with other ML-based techniques, demonstrating the utility of graph-based methods for capturing complex relational patterns in data.

Furthermore, the potential of graph-based clustering algorithms has not been thoroughly explored, even though clustering approaches have been included in a number of research. Our study aims to close this gap using various graph-based clustering algorithms. We employ ensemble clustering methods to improve the accuracy of detecting actors and events linked to different clusters.

The absence of attention to spatial patterns and the influence of various locations on armed conflict is another significant drawback of earlier research. By using clustering approaches to identify localities with varying death rates, our study closes this knowledge gap and provides law enforcement authorities with vital information to identify areas likely to see armed conflict ahead of time.

4 Methodology

Figure 1 illustrates the steps involved in implementing our proposed model, encompassing data collection and preprocessing. We detail the methodologies for constructing graphs, including the selection of attributes as nodes and the determination or construction of edge values from available data, which significantly influences the performance of graph-based algorithms. Subsequently, we outline the construction of graphs for clustering and regression-based approaches. We then demonstrate the processes for constructing and comparing clusters using both cluster-based methods and traditional KMeans clustering. Additionally, this section elaborates on how we assess effective values of cluster numbers and eccentricity for graph-based methods. Finally, we expound on the integration of GNN alongside ANN and traditional ML methods for fatality prediction.

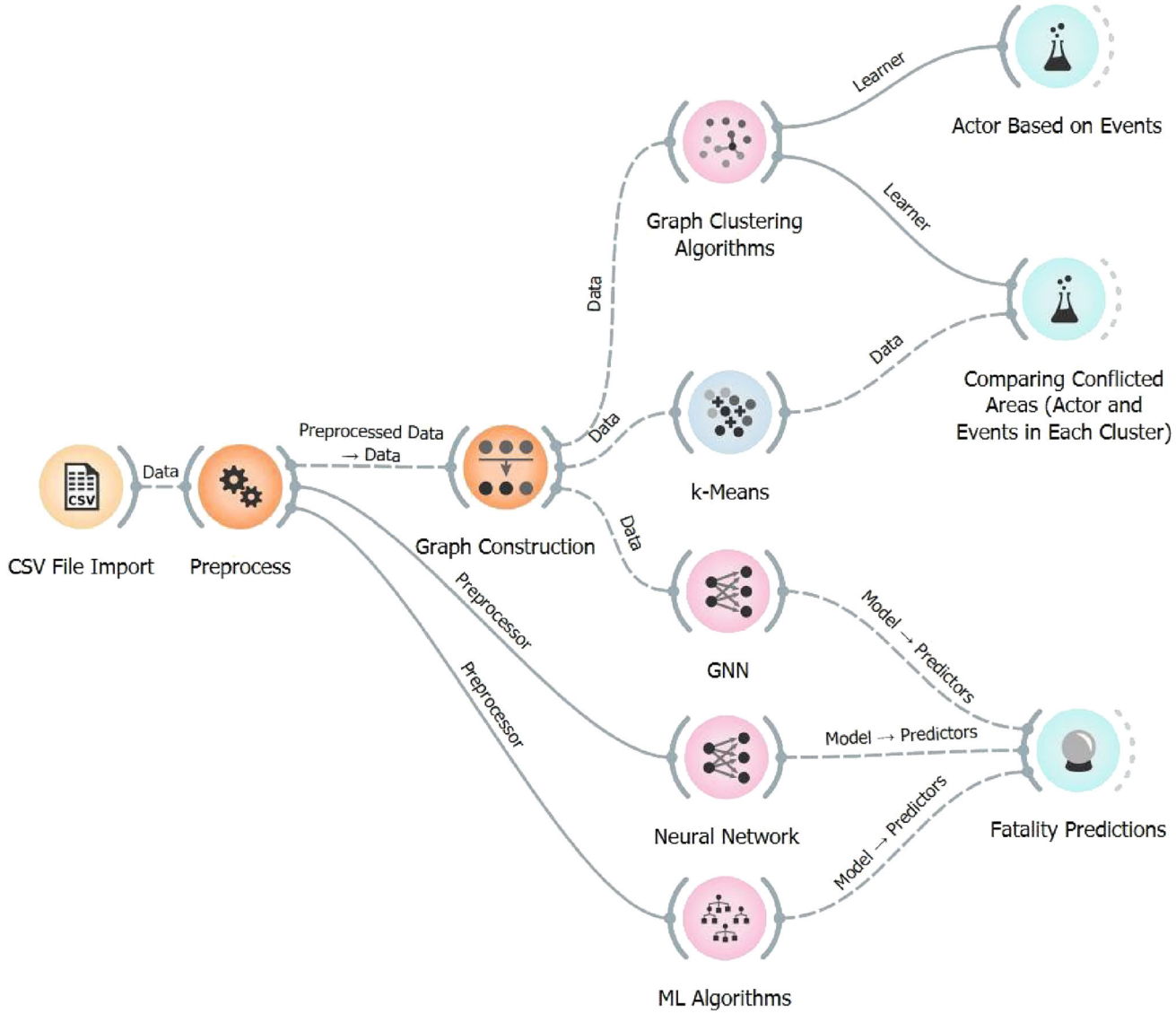

Fig. 1 Methodology diagram

4.1 Data collection and preprocessing

The dataset [26] was collected from the Kaggle website, which contains details of armed conflicts in Bangladesh from 2010 to 2021. Here, we have a dataset consisting of a total of 13,383 records. For the purpose of training in predictive modeling, we utilize 10,707 (80%) data points, while 2676 (20%) data points are reserved for testing. It has a total of 31 attributes [26], including actors, events, fatalities, location, and time, with other important features. For the preprocessing steps, we first removed the redundant attributes; as an example, we kept the value of administrative areas and removed the latitude and longitude columns that represent the same value. We then filtered the actors based on their counts. We considered that the actors participated in at least 193 events. The cut value 193 is decided based on the frequency of the actors, and the rest of the actors are addressed as "other." We also extracted months and years from another attribute "Event type" to analyze crime patterns over time. Missing values from different columns are replaced with the value "unknown." We used the one-hot encoding technique to convert the categorical features into numeric values. Moreover, we perform all basic preprocessing, such as removing duplicates and extraneous data points, to determine crucial preprocessed features.

There are several challenges to overcome when employing graph-based clustering techniques to analyze armed conflicts, particularly in the areas of graph construction and problem formulation.

Graph Construction: The precise and efficient creation of a graph is the first significant challenge. This involves determining the appropriate edges (such as interactions or links between actors/events) and nodes (such as actors or events). Given the complex nature of armed conflicts and the wide variety of actors and interactions, careful consideration must be taken to ensure that the graph appropriately depicts underlying facts. Furthermore, the performance and results of clustering analysis can be greatly impacted by the selection of the qualities to be used as nodes and edges.

Problem Formulation: Formulating the problem in a manner that maximizes the application of graph-based clustering approaches is another major challenge. This involves determining the criteria for grouping, such as similarity in behavior, location, or impact of those involved, as well as defining what, in the context of an armed conflict, defines a cluster. It can be challenging to apply typical clustering criteria and techniques without significant adaptation, owing to the inherent complexity and dynamism of armed conflicts. Generally, the comprehensive preprocessing and structuring of this dataset provides a solid foundation for using prediction models to examine patterns and trends in data on armed conflicts. These improved datasets can be used to improve the precision and effectiveness of the ensuing analysis.

4.2 Predicting fatality using graph neural network

In this section, we discuss how a graph neural network (GNN) is used to predict the fatality of armed conflicts in Bangladesh. We outline the features used, the construction of the adjacency matrix, and the GNN working mechanism for prediction.

4.2.1 Graph neural network

Graph neural network (GNN) is a deep learning architecture designed to process graph-structured data. GNNs are considered as a novel method of data modeling, especially in handling complicated data interactions, which sets them apart from other typical machine learning algorithms. GNNs use graph-based representations as compared to typical models that handle data points independently, enabling them to grasp complex relationships between data points. This feature is very helpful for modeling data, such as events in which interactions between entities are significant. For example, in our study on fatality prediction, we used a collection of complete features (event type, associated actors, interactions, and administrative details) for both classical machine learning models and GNNs. These features served as the basis for performance metric testing of more conventional models, such as support vector regression and gradient boosting regressors. The relationship context between data points may not be adequately captured by these models, because they handle each data point independently. However, under the same settings, GNNs showed remarkable performance. Because of their structure, GNNs can efficiently use relational information present in the graph format. The GNN's higher performance metrics make this evident; with the identical feature set but in a graph-based format, it achieved a remarkable R-squared score of 86%. Furthermore, even with a smaller feature set, the GNN showed notable improvement, demonstrating its adaptability and effectiveness when processing sparse data using graph-based techniques. For the most part, the key difference is that GNNs can accurately capture the complex correlations seen within event data because of their capacity to model data points in relation to one another. As opposed to conventional machine learning algorithms that see data points as separate entities, GNNs are especially well suited for situations where data interconnectedness is crucial (Table 2).

Table 2 Model performance metrics

Model	MSE	R2 score	RMSE
SVR	0.07	0.92	0.26
GradientBoostingRegressor	0.08	0.90	0.29
RandomForestRegressor	0.11	0.87	0.33
AdaBoostRegressor	0.11	0.87	0.33
DecisionTreeRegressor	0.09	0.89	0.30
KNeighborsRegressor	0.16	0.80	0.40

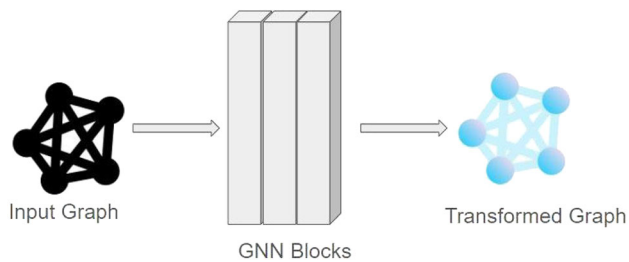

Fig. 2 Graph transformation using message passing

As stated earlier in the 3.4 section, we considered fatality prediction as an edge prediction problem. In edge-level tasks, the GNN predicts relationships (edges) between nodes by leveraging message passing to gather and aggregate information from neighboring nodes and update node embeddings accordingly.

Figure 2 depicts the graph transformation using message-passing technique. The model learns to capture graph connectivity patterns, thereby enabling accurate edge-level predictions in various applications. In a GNN, the node representations are updated using information gathered from nearby nodes. The GNN model creates a forward pass in this code that analyzes the node features (which represent

various characteristics of armed conflicts) and disseminates data across the graph. The GNN performs message passing in order to compile data from nearby nodes. The neighborhood connections between nodes are determined by the adjacency matrix, which is created using the k-nearest neighbors. The adjacency matrix's scipy sparse matrix format is transformed into a Deep Graph Library (DGL) graph object enabling effective computation during message transit.

ReLU activation is used between the two linear layers that constitute the GNN model. The node features are subsequently transferred through these levels during the forward pass. The node representations are updated by each layer, considering the data from nearby nodes. The nonlinearity of the model is introduced by the ReLU activation function, whereas the first linear layer modifies node characteristics. The output logits are produced by the second linear layer using the transformed input features. These logits represent the predictions generated for each node by the GNN model (i.e., the estimated number of fatalities per armed conflict).

4.2.2 Problem formulation

The GNN in this code examines graph data that represent armed conflicts, utilize message passing, and gather knowledge from nearby nodes to estimate the number of fatalities. The model is trained to improve its forecasts and discover edge forecasts (fatality counts) in the graph.

4.2.3 Dataset

The dataset included a variety of variables pertaining to armed wars in Bangladesh. We used 80% of our data for training, and the rest were utilized to test the model's performance. These characteristics are separated into numerical and categorical factors. The categorical features, such as "event type," "assoc actor 1," "assoc actor 2," and "admin1," are one-hot encoded to express them in a numerical manner. Using Standard Scaler, the numerical features were standardized. In the graph, these characteristics act as the node attributes.

4.2.4 Graph construction

A graph representation is necessary to use a GNN on a dataset. The connection between the graph nodes is described by the adjacency matrix. The adjacency matrix in this implementation was constructed using the k-nearest neighbor (KNN) technique. The algorithm uses a distance metric to determine the k-nearest neighbors of each node, and an adjacency matrix is then generated in response.

4.2.5 Implementation

The feedforward neural network employed in this investigation was the GNN model. It is composed of two linear layers, with ReLU activation placed in the space between them. The node features (categorical and numerical properties) propagated through the linear layers to create the output logits are the inputs to the GNN. As the objective is to estimate a continuous numerical value, the final layer of the GNN contains a single neuron to predict the fatality count. During training, the GNN optimizes its parameters to reduce the mean squared error (MSE) loss between the predicted fatality counts and the actual ground truth fatality counts. The Adam optimizer is used to update the model's parameters, and the training loss is monitored over the course of 50 epochs. Following training, the model was assessed using the R-squared score, as recommended by existing research, to determine the performance metrics of our predictive models, which measures how well the predictions of the model match the actual fatality counts in the test data. Common practices also led us to calculate losses using the mean squared error (MSE) and root-mean-squared error (RMSE), which form the basis of statistical analysis, offering a dependable way to evaluate model accuracy and being well known for their efficacy. The overall procedure is presented in Fig. 3.

4.3 Predicting fatality using machine learning models and artificial neural network

In this section, we explore the application of various machine learning (ML) methods along with artificial neural network (ANN) to predict the fatalities of armed conflicts from the dataset. As the target variable, fatalities were extracted from the armed conflict record dataset known as the ACLED. We employed GNN's same features as the GNN. Similarly, the categorical features were one-hot encoded in this case. A ratio of 80:20 was used to divide the dataset into training and test sets.

We trained and evaluated various regression models, including Linear Regression, ElasticNet, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, AdaBoost Regressor, Support Vector Regressor (SVR), and K-Nearest Neighbors Regressor (KNN). Each model was trained on the training set, and their performance was evaluated on the testing set. We measured the mean squared error (MSE), R-squared (R2), and root-mean-squared error (RMSE) as evaluation metrics. In addition to the ensemble methods, we also experimented with a simple artificial neural network (ANN) architecture. The model is an artificial neural network (ANN), and its hidden layer sizes = (300, 150) specification specifies that it has two hidden layers, each with 300 and 150 neurons. The hidden layers are activated using the ReLU activation function (activation =

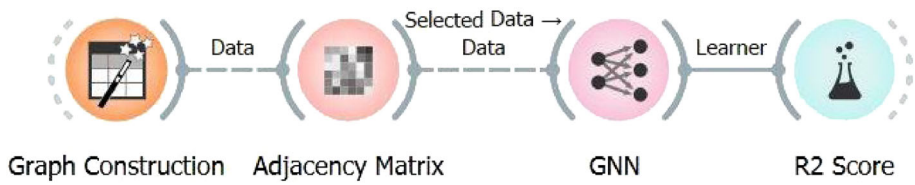

Fig. 3 Graph construction for clustering areas

Fig. 4 Graph construction for clustering actors

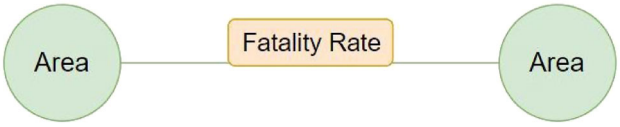

Fig. 5 Graph construction for clustering areas

"relu"), which introduces nonlinearity and enables the model to recognize intricate patterns in the data. The "Adam" solver is utilized during training and adjusts the learning rate and optimizes the weights of the model to achieve quicker convergence and better performance on the supplied dataset.

4.4 Graph-based clustering

4.4.1 Graph construction

An important step in this study is to convert the data into graphs so that we can apply different clustering algorithms to compare and analyze the results of traditional clustering and machine learning models. We have taken proper care to construct graphs for both clustering and fatality prediction because the accuracy of graph algorithms depends mostly on the structure of the graph. From our methodology diagram in Fig. 1, we can see that we need to construct graphs for both clustering and predicting fatalities, which is a regression-based analysis. We constructed a graph using different approaches to address both scenarios. To cluster the actors, we used the attributes "Actor1" and "Actor2" as vertex and "Events" as their edges. In simple terms, when two actors are involved, the kind of event associated with them works as an edge value which is depicted in Fig. 4.

When deciding the cluster upon areas, we take "Admin1" which indicates the administrative area (Divisions) as a vertex and the difference of fatality rate between them works as edge value that is presented in Fig. 5. The fatality rates have been defined as follows: First, data related to fatalities and the corresponding administrative areas were extracted.

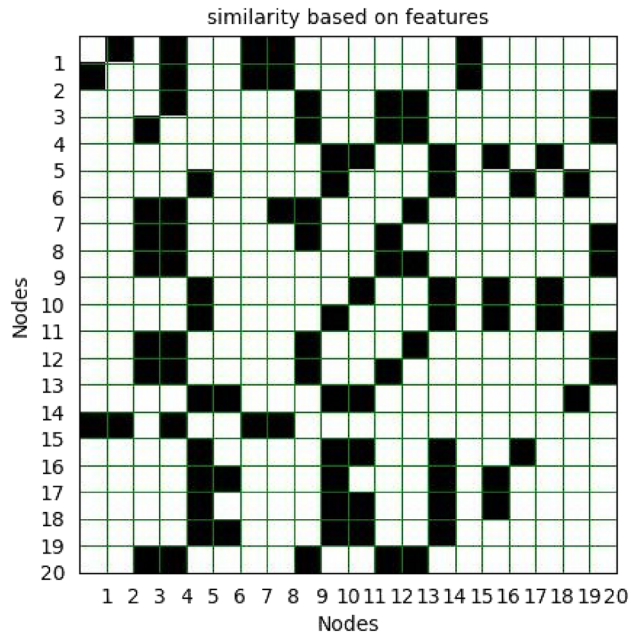

Fig. 6 Adjacency matrix using KNN considering 20 nodes

To provide an overview of typical fatality scenarios, the average fatality rates were then computed for each location by dividing the total number of fatalities by the number of occurrences or periods recorded. These averages are used to create a graph where each area is represented as a node, and the size and color of the node signify the severity of fatalities. Later, to assist in identifying patterns and clusters of locations with similar risk profiles, a similarity matrix measures how closely mortality rates vary across areas.

Besides clustering, for the predictive modeling graph construction, we selected the following features ["event type," "assoc actor 1," "assoc actor 2," "interaction," "admin1," and "fatalities"]. Each row of the selected features functions as a node. The connection between nodes is determined by calculating the Euclidean distances between node features. We considered the five nearest neighbors from the adjacency matrix that had the highest similarity based on the features which is shown in Fig. 6. By constructing a graph, we created an edge prediction problem to predict fatalities.

4.4.2 Graph-based cluster of actors on events

First, we try to group actors according to how they interact with each other. Several clustering algorithms receive the graph produced by the previous layer, which has actors as nodes and events as edge values. Although there are many graph clustering techniques available, we only choose those that are relevant to our area of study. To accomplish this clustering goal, we evaluate the suitability of various techniques, including "Spectral clustering," "Girvan-Newman," "Markov Clustering," "KMeans," and "DBSCAN."

4.4.2.1 Spectral clustering Is a graph-based clustering technique that uses the eigenvectors of the Laplacian matrix to embed the data into a lower-dimensional space and then applies standard clustering algorithms to group the points. Using the eigenvectors of a similarity matrix created from the data, spectral clustering is a potent technique that divides data points into clusters. Spectral clustering is especially helpful for datasets with complicated structures or variable cluster densities since, in contrast with conventional methods like KMeans, it does not need predetermining the number of clusters. To begin, the algorithm creates a graph representation of the data, in which data points are represented by nodes, and pairwise similarities or affinities between points are indicated by edges.

4.4.2.2 Girvan–Newman Is a hierarchical agglomerative clustering method for graphs, based on removing edges with the highest betweenness centrality to gradually break the graph into connected components, resulting in a dendrogram structure that can be cut at different levels to obtain clusters. A hierarchical agglomerative technique for community detection in networks is the Girvan–Newman algorithm. The program gradually disassembles the network into smaller components, exposing the underlying community structure, by repeatedly eliminating these edges. The Girvan–Newman algorithm can identify communities of different sizes and densities inside a network and does not require the number of communities to be specified in advance.

4.4.2.3 Markov Clustering Algorithm (MCL) Is an iterative graph clustering technique based on the simulation of random walks in a graph. It uses matrix multiplication and inflation to cluster nodes with high probabilities of being in the same cluster. Specifically created for graph-based data, the Markov Clustering Algorithm (MCL) is an unsupervised clustering technique. Through the simulation of random walks, it turns the graph into a Markov chain. MCL simulates the processes of diffusion and contraction inside the graph by performing steps of inflation and expansion iteratively. As a result, dense clusters are formed, with low inter-cluster connectivity and strong intra-cluster connectivity among the nodes. MCL has been widely utilized in biological network research, social network analysis, and web data mining applications. It is especially useful for finding clusters in complex networks.

4.4.2.4 KMeans Is a centroid-based clustering algorithm that partitions data into K clusters by iterative assigning each data point to the nearest cluster center (centroid) and then updating the cluster centers based on the mean of the points in each cluster. It divides data points into a fixed number of groups according on how similar they are. The method involves repeatedly allocating data points to the closest centroid and adjusting the centroids to reflect the average of the data points allocated to every cluster. This process keeps going until either the convergence requirement is satisfied or the centroids stabilize. Because to KMeans's scalability and simplicity, it is a popular and computationally efficient method. But it is sensitive to initialization and necessitates predetermining the number of clusters, which frequently leads to less-than-ideal solutions. KMeans is used in many different domains, including document categorization, picture compression, and consumer segmentation, despite its drawbacks.

4.4.2.5 DBSCAN Is a density-based clustering algorithm that groups together data points that are close to each other in the feature space, and it can identify outliers as noise points. It does not require specifying the number of clusters in advance. It clusters together densely packed data points according to their density. In contrast with KMeans, DBSCAN may discover clusters of any shape or size and does not require a prior definition of the number of clusters. It functions by specifying two parameters: minPts, the bare minimum of points needed to form a dense zone, and epsilon (), the radius around which neighboring points are regarded as belonging to the same cluster. DBSCAN uses the density of the dots in relation to these factors to classify them as core, border, or noise.

After applying different graph clustering algorithms to actions and analyzing them, we used an ensemble method to effectively combine the various results produced by these approaches. Each algorithm was selected based on its unique strategic handling of the complicated actor–event network data. Using the distinct advantages of each algorithm, such as spectral clustering, Girvan–Newman, MCL, k-means, and DBSCAN, this ensemble technique guarantees the generation of strong clusters. When both approaches are combined, it is possible to produce a more accurate and subtle actor grouping in conflict analysis, as demonstrated by the combined results shown in Table 3. Our findings can be more effectively interpreted and applied to comprehend the dynamics of armed conflicts thanks to this integrative approach.

Table 3 GNN details

Parameter	Description
Activation function	Relu
Learning rate	0.01
Number of hidden layers	300
Optimization method	Adam

4.4.3 Graph-based cluster of areas and comparisons

In the second clustering approach, we have clustered the Divisions based on the fatality rate among them. Areas can be detected with different clustering algorithms but we take into consideration of spectral clustering which is a graph-based algorithm and traditional k-means which is a machine learning-based algorithm. The reason is they are closely related and we want to examine the similarities of the cluster-constructing nature of these two different approach algorithms. Spectral clustering embeds data into lower dimension and then uses the k-means type algorithm internally. The comparison and hidden patterns of results are discussed in broad in the "Results and Discussion" section.

5 Results and discussion

This section describes the results of the different graph-based clustering algorithms for clustering actors based on events and areas. Moreover, it explains how the cluster number is determined, and how the results are compared with those of traditional clustering algorithms. Subsequently, we discuss the performance and effect of selecting different features that affect the results for graph neural networks, artificial neural networks, and machine learning algorithms.

5.1 Fatality prediction and comparative analysis of different models

In this section, graph-based neural networks, artificial neural networks, and conventional machine learning models were compared for their ability to predict deaths. We also discuss how feature reduction affects the model performance, and how our findings are relevant. For each model, we first use the feature set ["event type," "sub event type," "assoc actor 1," "inter1," "assoc actor 2," "inter2," "interaction," "admin1," and "fatalities"]. In Tables 3 and 4, we have listed the details parameter used in model train for GNN and traditional machine learning, respectively.

Table 4 Machine learning details

Algorithm	Default parameters
Support Vector Regression (SVR)	- kernel: 'rbf',—C: 1.0,—epsilon: 0.1,—gamma: 'scale',—shrinking: True
GradientBoostingRegressor	- loss: 'ls',—learning_rate: 0.1,—n_estimators: 100,—subsample: 1.0,—min_samples_split: 2,—min_samples_leaf: 1,—max_depth: 3
RandomForestRegressor	- n_estimators: 100,—criterion: 'mse',—max_depth: None,—min_samples_split: 2,—min_samples_leaf: 1,—max_features: 'auto',—bootstrap: True
AdaBoostRegressor	- base_estimator: None,—n_estimators: 50,—learning_rate: 1.0,—loss: 'linear'
DecisionTreeRegressor	- criterion: 'mse',—splitter: 'best',—max_depth: None,—min_samples_split: 2,—min_samples_leaf: 1,—max_features: None
KNeighborsRegressor	- n_neighbors: 5,—weights: 'uniform',—algorithm: 'auto',—leaf_size: 30,—p: 2,—metric: 'minkowski'

5.2 Performance of traditional machine learning models

We first evaluated several traditional machine learning models, including Support Vector Regression (SVR), GradientBoostingRegressor, RandomForestRegressor, AdaBoostRegressor, DecisionTreeRegressor, and KNeighborsRegressor, using the selected features. The performance metrics for each model are summarized in Table 2.

5.3 Graph-based neural networks performance

Afterward, we assessed how well graph-based neural networks (GNNs) predicted deaths by utilizing the selected features. Using a GNN model with 300 hidden layers and 50 epochs produced an outstanding 86% R-squared. An accuracy rate of 80% was also obtained with artificial neural networks (ANNs) using a similar architecture. This finding highlights how much better GNNs are than traditional machine learning models in capturing the complex correlations present in event data, which leads to forecasts of fatalities that are more accurate.

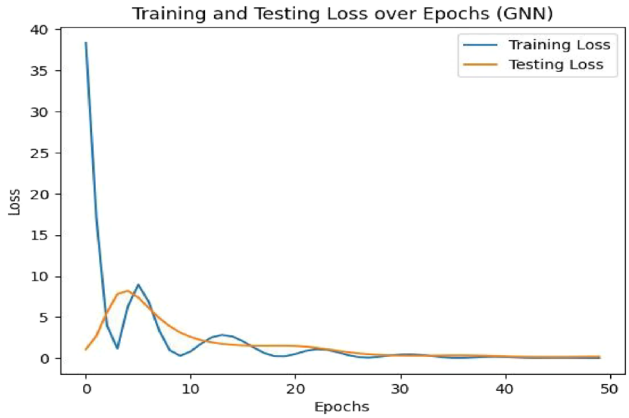

Fig. 7 Loss curve of GNN

Table 5 Actor clusters

Cluster 1	["Civilians (Bangladesh)," "other," "Police Forces of Bangladesh (2009-) Rapid Action Battalion," "Police Forces of India (2004–2014) Border Security Force"]
Cluster 2	["unknown," "Police Forces of India (2014-) Border Security Force," "AL: Bangladesh Awami League," "BNP: Bangladesh Nationalist Party," "ICS: Islami Chhatra Shibir"]
Cluster 3	["BCL: Bangladesh Chhatra League," "JCD: Bangladesh Jatiotabadi Chatra Dal," "Jel: Jamaat-c-Islami"]
Cluster 4	["Police Forces of Bangladesh (2009-)," "Rioters (Bangladesh)," "Unidentified Armed Group (Bangladesh)," "Jubo League"]

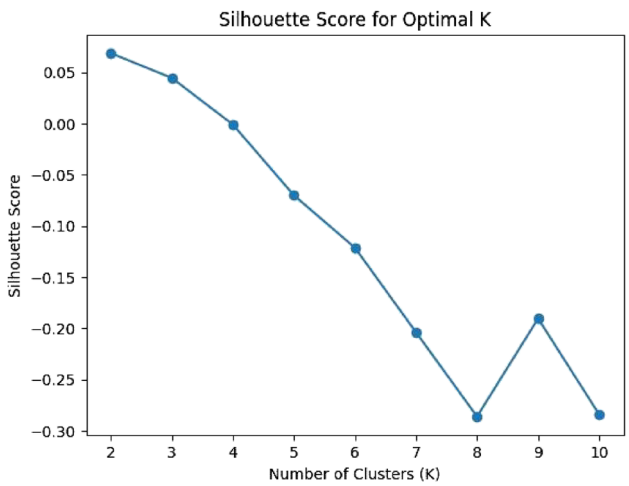

Fig. 8 Silhouette score for spectral clustering

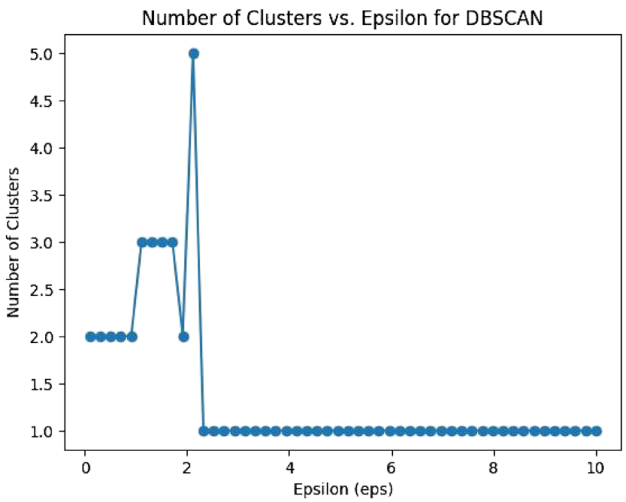

Fig. 9 Epsilon score of DBSCAN

5.4 Impact of feature reduction in performance

To understand the impact of feature reduction on model performance, we further investigated a reduced feature set consisting of only two features: "event" and "admin1 (administrative Area)." With the reduced feature set, the ANN model achieved a significant increase in accuracy, with an R-squared score of 90%. Surprisingly, the GNN model exhibited an even more substantial improvement with the reduced feature set, achieving an impressive R-squared score of 93%. The loss curve of GNN that is presented in Fig. 7 verifies the improved performance of GNN.

This outcome emphasizes the ability of the GNN to exploit graph-based representations and effectively capture relevant patterns from a reduced feature space. The GNN's ability to leverage graph-based representations allows it to capture complex interdependencies among different events and features, making it a promising choice for scenarios with resource constraints or sparse datasets.

5.5 Result of clustering actors on events

In this study, we examined how our recommendations for actors were clustered based on event types. Our toolkit for clustering graphs includes Girvan–Newman, spectral clustering, k-means, Markov clustering (MCL), and DBSCAN. In contrast with spectral clustering and k-means, which used Silhouette score presented in Fig. 8 and the elbow method for the best K determination, respectively, Girvan–Newman gave a hierarchical perspective on community structures. Table 5 provides insights about the clustering based on the event type. Compared to DBSCAN, the Markov Clustering Algorithm (MCL) revealed clusters based on probabilistic flows. Spectral, Girvan–Newman, and DBSCAN suggest that the optimal value of cluster number is 2 which is shown in Fig. 9 whereas k-means and Markov clustering suggest eight different clusters.

We use an ensemble method to efficiently aggregate the results of these algorithms and produce strong clusters of

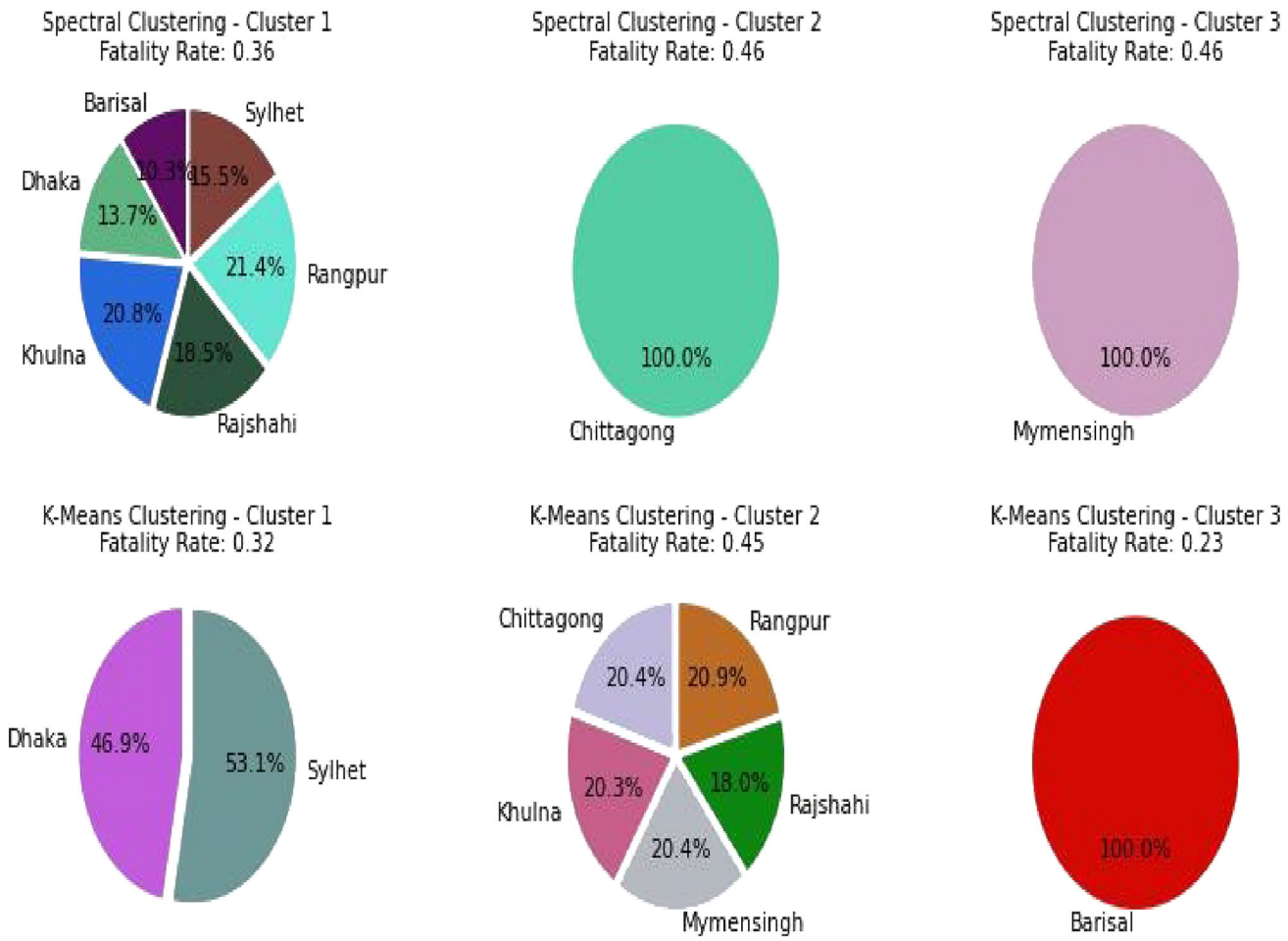

Fig. 10 Comparison of clusters based on areas

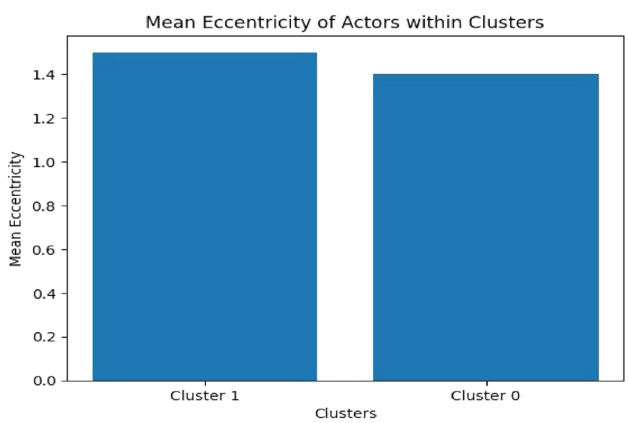

Fig. 11 Eccentricity of actors from spectral clustering

actors because each of these clustering algorithms uses a different strategy that can produce a variety of clustering outcomes. The results of Girvan–Newman, spectral clustering, k-means, Markov clustering (MCL), and DBSCAN can all be used to improve the final actor clusters for our recommendation system. By doing so, we can take advantage of the advantages of each approach.

5.6 Clustering conflicted areas

In this section, we examine the findings of the application of two clustering techniques, spectral clustering and k-means clustering, to an area-based dataset. The dataset included data from different administrative regions (admin1) and associated average fatality rates. Based on the frequency of fatalities, these locations were grouped into discrete clusters. First, we used spectral clustering, a graph-based clustering algorithm. Each admin1 region is represented as a node in the graph that we created, with the similarity between nodes determined by the absolute difference in fatality rates. Similarly, we immediately applied the similarity matrix to the conventional k-means clustering algorithm. The ideal outcome of K for the spectral and k-means methods, respectively, is 2 and 3, obtained by utilizing the silhouette score and elbow method. To make the comparisons between the two

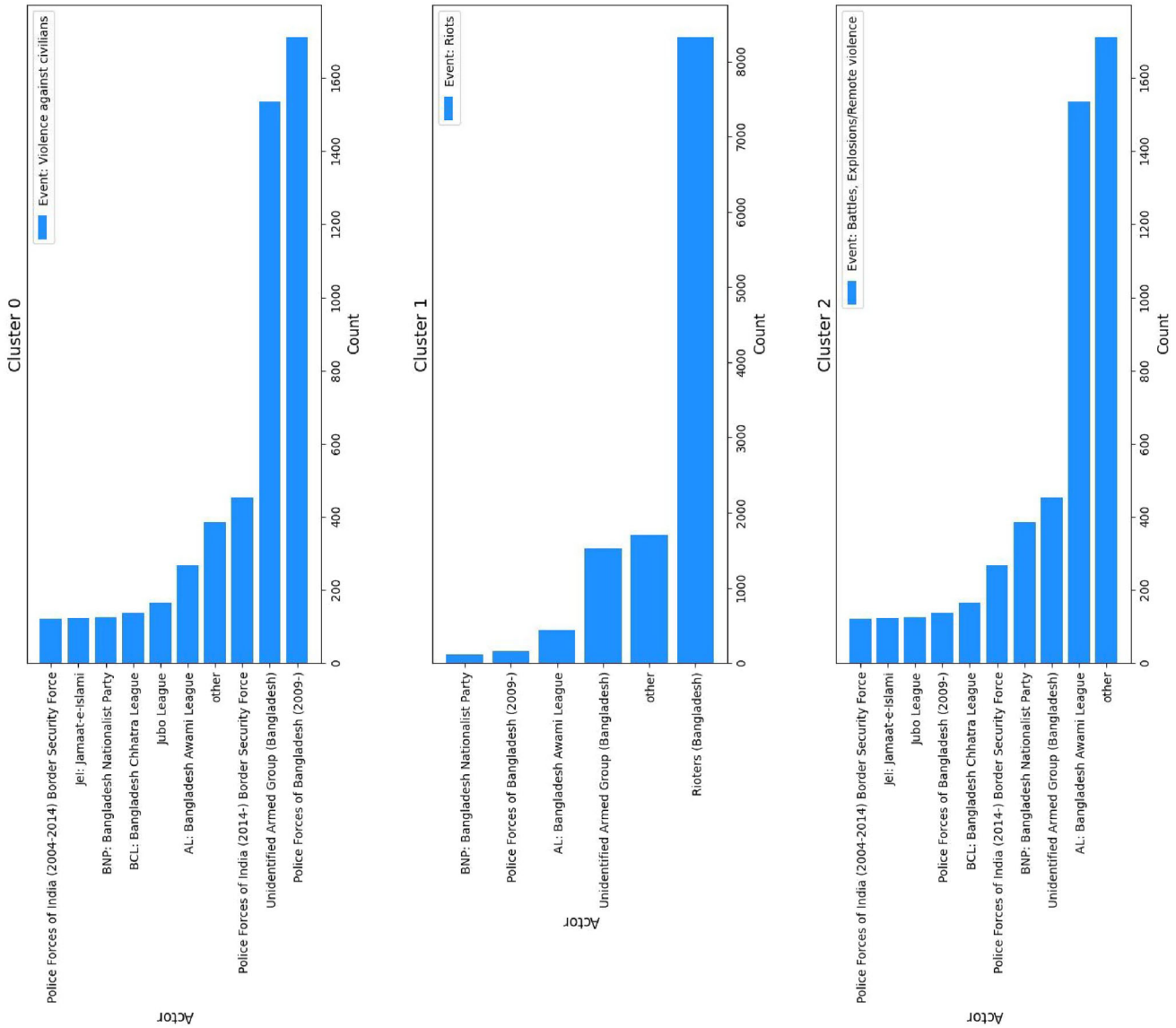

Fig. 12 Events and actors in each cluster

better, we purposefully used three clusters. Using clustering with three clusters, we were able to generate labels for each region that indicated their participation in the appropriate clusters. Subsequently, using both techniques, we calculated the average fatality rate for each cluster. Clustering findings were visualized using pie chart. The pie chart shows in Fig. 10 which represents the percentage of each area's fatalities attributable to each cluster, with the average fatality rate for each cluster (Fig. 11).

5.7 Events and associated actors in the clusters

As we identified the clusters of areas with fatality rates, now we can answer the questions like "In the Highest fatality-prone areas which actors were involved?" or "What are the events associated with those regions?" This helps us to identify the hidden factors that fuel crimes. Law enforcement agencies can take advanced precautions and preventive measures by analyzing these factors. We used k-means clustering to find groupings of events and actors based on their similarities because the workings of the spectral and k-means algorithms are essentially identical. It sorts events and players into discrete clusters by building a graph with event-related nodes and using k-means to reveal their relationships and patterns. To help comprehend the relationships and characteristics of the events and actors within each cluster, the results are summarized in Fig. 12 while also listing their names.

We evaluated the actors that were part of the clusters for eccentricity. Eccentricity is a metric used to analyze a graph's structural characteristics and connections between its nodes.

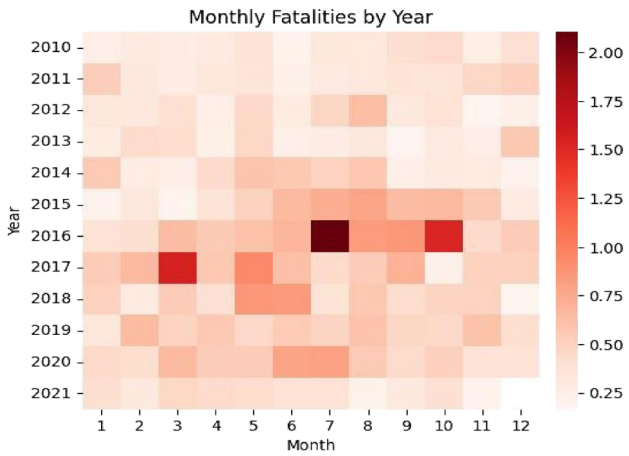

Fig. 13 Heatmap of fatality over time

In the graph, nodes with lower eccentricities typically had more connectedness. Nodes that are grouped together in the context of k-means clustering are not directly connected to one another or are part of several graph segments. Therefore, instead of k-means clustering, we use spectral clustering to calculate the eccentricity. Figure 8 demonstrates that the cluster number was 2. Figure 11 displays the eccentricity of the actors in the clusters, and eccentricity has a lower score, indicating a strong relationship between the actors within the clusters.

5.8 Knowledge graph by decoding temporal violent incidents

To comprehend the links and temporal patterns among the various entities involved in violent incidents across time, an interactive knowledge graph must be created using the available data. First, the data displayed in Fig. 13 were used to compute the average monthly number of fatalities for each year, emphasizing the significance of timing in influencing the level of violence.

In the knowledge graph, objects, such as actors and locations, are represented as nodes, and their interactions are shown as edges. A more in-depth investigation of temporal trends is based on the graph nodes' associations with characteristics such as the number of events and average mortality. Nodes can also show which actors are present in the locations during which months. Users can dynamically explore the knowledge graph and acquire insights into how violent behavior changes over time by utilizing interactivity. When viewed in the context of particular months or years, it enables viewers to find connections between actors, places, and happenings.

Understanding how violence changes over time critically depends on time. The most violent months can be examined, actor involvement tendencies can be found, and areas with high incidence rates can be located during particular periods. A sample of the knowledge graph is shown in Fig. 14. Users can identify new patterns and gauge the success of actions over time as the knowledge graph dynamically refreshes with fresh data. Overall, the interactive knowledge graph provides a strong tool for visualizing, investigating, and understanding the temporal dynamics of violent episodes, enabling strategic responses to violence and its root causes.

6 Conclusion and future work

This study conducts a comparative analysis between graph-based clustering and traditional clustering methods to model armed conflicts in Bangladesh and compares the performance of graph neural networks (GNNs) against conventional machine learning (ML) algorithms, particularly artificial neural networks (ANNs), to predict fatalities. This study aims to understand armed conflict dynamics and forecast mortality rates, which are the key components of conflict research. Graph-based clustering algorithms capture the intrinsic relational structure of contradictory data, whereas classic clustering methods focus on feature engineering and distance measurement. The findings highlight the usefulness of graph-based clustering techniques in providing insights into the dynamics and structure of violent conflicts in Bangladesh. Additionally, the comparison demonstrates that GNNs outperform some conventional ML techniques in predicting lethality, underscoring the importance of understanding the relational relationships within the data. Overall, the study shows the utility of graph-based methodologies and GNNs in comprehending and forecasting patterns of armed conflict. Graph-based clustering approaches offer valuable insights into complex data structures, yet they possess several limitations. The datasets with few occurrences that could affect the prediction were filtered out. These biases within datasets can distort graph representations, impacting clustering outcomes. Skewed or incomplete data, along with outliers, can compromise the accuracy of the clustering algorithm. We also worked on a selected feature set. High-dimensional or nonlinear data can pose challenges for graph construction and parameter specification, hindering the effectiveness of graph-based clustering. Furthermore, we made an effort to reduce some computational complexity by utilizing KNN to compute the adjacency matrix while only taking 20 nodes into account. Computational complexity presents scalability issues for large datasets, while traditional graph construction methods may struggle with identifying clusters of varying densities or irregular shapes.

In the future directions, some additional data sources, such as social media and satellite imagery, can be integrated to enhance the accuracy and granularity of conflict

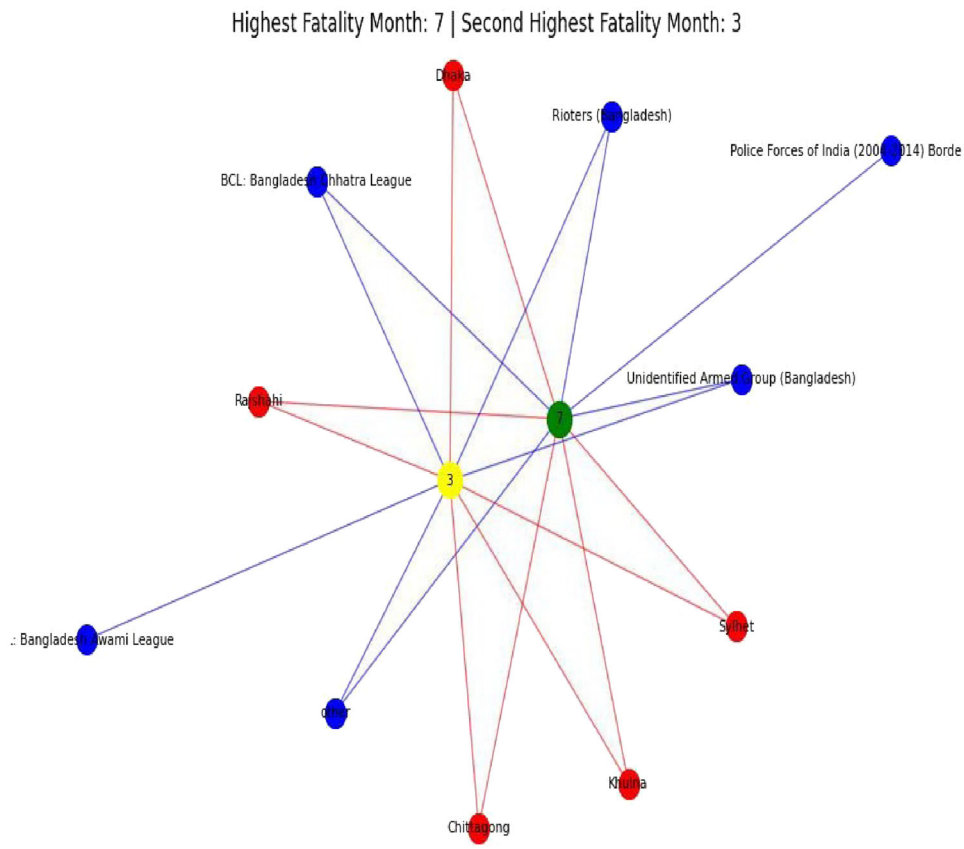

Fig. 14 Building a knowledge graph

analysis. Also, investigating alternative graph-based modeling techniques, such as community detection algorithms like Louvain algorithm, Info map algorithm, Label propagation algorithm, etc., could offer new insights and potentially improve clustering accuracy. Lastly, applying the methodology to other contexts of conflict, such as social unrest or political instability, could broaden the scope of the research and uncover common patterns or underlying structures across different conflict scenarios. The knowledge collected from this research can help in designing more reliable conflict analysis models and tactics for conflict prevention and resolution in Bangladesh.

Authors' contributions D.N.S, S.P.S, M.M.H, and M.A.R wrote the main manuscript text. S.P.S, M.M.H, and M.A.R did the experimentation and data analysis part. D.N.S. coordinated the whole study and reviewed the manuscript. All authors analyzed the data, discussed the results, and approved the final manuscript.

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Declarations

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Ethical approval There are no studies by any of the authors in this article that used humans or animals as subjects.

Consent to participate Consent to publish.

Consent for publication All authors agreed to publish.

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